Alireza Kaviani

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Sharif University of Technology, Tehran, Iran

Education

Sharif University of Technology September 2022 – July 2026
Bachelor of Science in Computer Engineering

Allame Helli High School July 2019 – July 2022
Focus on Informatics Olympiad

Research Experience

Tehran Institute for Advanced Studies (TeIAS) January 2023 – Present
Advisor: Prof. Masoud Seddighin

University of Maryland June 2024 – Present
Advisor: Prof. MohammadTaghi Hajiaghayi

Max Planck Institute for Informatics July 2025 – December 2025
Advisor: Prof. Danupon Nanongkai

Publications

[1] Iman Gholami, Mohammad T. Hajiaghayi, Mahdi JafariRaviz, **Alireza Kaviani**, Soheil Mohammadkhani, and Amir-Mohammad Shahrezaei. Constructive Arboricity in Almost-Linear Time for Dense Graphs. *Submitted to STOC 2026*. arXiv:2610.13937

[2] Mohammad T. Hajiaghayi, Mahdi JafariRaviz, **Alireza Kaviani**, and Soheil Mohammadkhani. Breaking the Exponential Barrier: The First Polynomial-Time Algorithm for the Győri-Lovász Theorem. *Submitted to STOC 2026*. arXiv:2608.30945

[3] Ehsan Heidari, **Alireza Kaviani**, Masoud Seddighin, and AmirMohammad Shahrezaei. Improved Maximin Share Guarantee for Additive Valuations. *Proceedings of the 37th ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2026. arXiv:2510.10423

[4] **Alireza Kaviani**, Alireza Keshavarz, Masoud Seddighin, and AmirMohammad Shahrezaei. Improved Approximate EFX Guarantees for Multigraphs. *Proceedings of the 21st Conference on Web and Internet Economics (WINE)*, 2025. arXiv:2506.09288

[5] **Alireza Kaviani**, Masoud Seddighin, and AmirMohammad Shahrezaei. Almost Envy-Free Allocation of Indivisible Goods: A Tale of Two Valuations. *Proceedings of the 20th Conference on Web and Internet Economics (WINE)*, 2024. arXiv:2407.05139

Honors & Awards

- **International Olympiad in Informatics (IOI)**: Gold Medal (Rank 12, 2022); Silver Medal (2021)
- **Iran National Olympiad in Informatics (INOI)**: Gold Medal (2020, 2021)
- **ICPC West Asia Regionals**: Champion (2025), Gold Medal (2022, 2023, 2024)
- **Codeforces**: Legendary Grandmaster (Peak Rating: 3012)

Research Interests

Graph Algorithms, Complexity Theory, Algorithmic Game Theory.

Skills

Programming: C++, Python, NumPy
Tools: Git, LaTeX
Languages: Persian (Native), English (Professional Proficiency)